Assessing Factors Influencing the Reliability of Smart Insole Measurements

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***Abstract*— Smart insoles are widely used in Human Activity Recognition research, in areas such as health and sport. The majority of smart insole research so far has tended to focus on the intellectual frontier of what is possible, and less on how to turn that into reliable, market ready products. This paper looks at the reliability of smart insole measurements. It describes experiments that illustrate some of the limitations on the accuracy and robustness of smart insole measurements. In particular, it highlights that any application that depends on the magnitude or pattern of pressure or inertial sensor measurements across sessions and between different subjects will be highly dependent on consistent and accurate fitting of the smart insoles. Smart insoles have strengths and weaknesses, and this paper gives examples of applications which play to smart insoles' strengths, as well as those where greater care needs to be taken to work around the weaknesses.**

Keywords—Smart insoles, reliability, gait measurement, human activity recognition.

I. Introduction

Human Activity Recognition (HAR) is an area of research looking at how to determine types of human activity using data from sensors. The results of this research can be applied in a wide variety of areas, including health, sport, surveillance and entertainment [1].

Several different approaches to HAR have been investigated including vision-based systems, sensors in devices the human subject wears and sensors in the devices the human subject interacts with [2]. The best approach depends on many factors including the area of application, accuracy required, accessibility, time available and cost. Each approach has strengths and weaknesses. Wearable devices have grown in popularity in recent years as the variety of devices available has grown, and the devices themselves have become cheaper and less intrusive. One type of wearable device that is showing potential is the smart insole.

Many studies can be found in the literature that use smart insoles for HAR [3]. These studies typically achieve high accuracy but to be useful in real world applications the models need to be based on reliable insole measurements. Limited research has been reported on assessing the reliability of smart insole measurements. There is no single definition of the term "reliable". The "Research Methods Knowledge Base" [4] identifies four general classes of reliability estimates. This study investigates smart insoles within the context of one of these classes – Test-Retest reliability. This can be defined as the consistency of a measure from one time to another. In particular, this study looks at whether smart insole measurements are accurate and repeatable over time and in different conditions, and highlights factors that may impact accuracy and repeatability.

The remainder of this paper is organised as follows. Section II outlines related work in the general area of HAR using Smart insoles, and the specific area of reliability. Section III then describes the methodology used and Section IV describes the experiments and results carried out. Based on these findings, Section V outlines the reliability implications of using smart insoles in HAR applications and gives some examples where smart insoles are well suited, and others where extra care is needed in design and execution. Section VI lists the conclusions and suggestions for further work.

II. Related Work

This section gives a high level overview of HAR studies carried out using smart insoles. It then looks at the steps needed to go from a research project to a market ready solution, and finishes with the key role that smart insole measurement reliability plays in this process.

A. HAR Studies using Smart Insoles

HAR studies can be categorized by type of activity, e.g.

1. Identifying common physical activities, such as walking, sitting, standing, climbing stairs [5]. This could be useful for monitoring the elderly, where unexpected behaviour (or lack of behaviour) could indicate a problem or accident.
2. Detecting the presence or progression of a medical condition, or for rehabilitation, e.g., the detection of Parkinson's Disease [6].
3. Monitoring sports performance and the effectiveness of training techniques [7].

B. Steps to Develop a HAR System

There are five key steps to developing a smart insole based HAR system.

1. Decide whether to buy a commercially available smart insole system (such as the ActiSense Kit used in this study [7]), or develop a new system from scratch with specific features (e.g., Tana et al [8] presented a low cost system built using piezo-resistive material).
2. Decide what to measure. This is more difficult than it sounds as there are few widely agreed features in many areas, e.g., for neurodegenerative diseases, Tăuţan et al [9] proposed a taxonomy of the specific clinical features of the most common neurodegenerative diseases. The list of features revealed a lot of overlap, and hence the difficulty in diagnosing a specific disease and measuring progression.

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3. Decide how to measure. For any particular feature, there are often a wide variety of completing solutions, e.g. looking again at neurodegenerative diseases, Tăuțan et al [9] also proposed a taxonomy of computational approaches, listing the clinical features and what sensors can be used to measure them for diagnosis, classification and progression monitoring, e.g., accelerometers, gyroscopes and force sensors can be used for measuring gait.
4. Decide how to assess the effectiveness of the measurement. Again, there are a wide variety of methods and "there is no consensus on evaluation metrics" [9].
5. Turn a basic research project (that demonstrates what is possible) into a robust implementation. For example, for a medical scenario, Kubota et al [10] suggested this requires clinometric validation for regulatory approval, demonstrating that –
   - The solution provides an accurate parameter of a clinically relevant feature of the disease;
   - There is confirmed evidence that this parameter has an ecologically relevant response within some specific clinical application;
   - A target numerical range exists in which the parameter measures adequate treatment response.

The reliability of smart insole measurements is relevant to all these steps, in particular to help decide how best to measure some feature of an activity, how best to assess the accuracy of the measure, and how to demonstrate that the measurement can detect change over time.

C. *Reliability of Smart Insole Measurements*

There are a number of general challenges with smart insole measurements, e.g., data quality issues, data drift and non-stationarity [10]. The impact of these can be reduced by using simpler ML algorithms that are less likely to be brittle in the face of changing distributions [10] (although possibly at the expense of accuracy).

Other papers identify specific point solutions that exploit some reliable aspect of a particular feature, e.g., Hundza et al [11] suggest using gyroscope angular rate reversal for identifying the start of strides in walking, because the change is rapid and can be identified relatively easily in many different types of stride pattern.

While efforts have been made to improve reliability, the majority of the research so far has tended to focus on "the intellectual frontier of what is possible (c.f. basic research) and not on how to turn that into robust implementation (c.f. translational research)*"* [10].

III. METHODOLOGY

This section describes the smart insole kit and how it was used in this study, and defines some of the terms used later.

A. *The Smart Insole Kit*

The specific smart insoles used in this study are an ActiSense Kit from IEE Luxembourg S.A. [7]. The kit has two main parts (Fig. 1).

Fig. 1. The ActiSense Kit, showing the 8 pressure sensors (below the red layer) and the ECU (top right). (Picture © IEE Smart Sensing Solutions)

1. An insole containing eight pressure cells, located where the foot to ground impact is likely to be highest (heel left, heel right, arch, met-1, met-3, met-5, toes and hallux, Fig. 2).
2. An Electronic Control Unit (ECU) that includes an accelerometer, a gyroscope and a magnetometer, each with three degrees of freedom. The ECU also stores the data collected by the sensors and supports transfer of the data via Bluetooth to other devices. The ECU clips to the side of the shoe as shown.

There are 17 sensors per foot (8 pressure sensors plus 9 inertial sensors), giving a total of 34 sensors. The frequency at which sensor readings can be recorded is configurable. All the experiments carried out as part of this investigation used a sampling frequency of 25Hz

B. *Experimental set-up and Terminology*

A single adult male was used for all the experiments carried out in this study. XL insoles were used (suitable for shoe size EU 44.5-47 [7]), along with size 46 shoes.

The ActiSense smart insoles each contain 8 individual high dynamic pressure cells. [7] states that pressure "data is recorded with a 12bit resolution (0 – 4095)", but doesn't indicate how these data can be converted into an actual pressure or force. In this study, the pressure measurements are simply reported as a number without units.

The following terms are used in this study –

- Total pressure – the sum of the 16 pressure readings across both insoles in a sample.
- Left/Right foot pressure – the sum of the 8 pressure readings from the left or right insole.

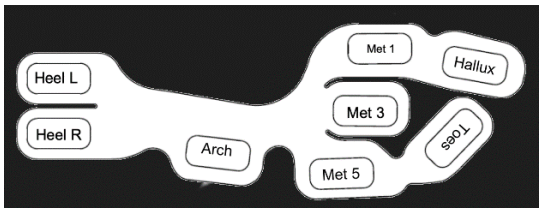

Fig. 2. The right smart insole showing the position of the 8 pressure sensors.

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- Front/Rear pressure – the sum of the 10 sensors at the front the insoles (Hallux, toes, met-1, met-3 and met-5) or the sum of the 6 at the rear (arch, heel-l, heel-r).
- Average total pressure – the average of the total pressure across multiple samples, typically a whole number of complete strides.

IV. Experiments and Results

This section outlines the experiments carried out and the results obtained. They are split into those related to the pressure sensors, the inertial sensors and the overall system.

A. *Pressure Sensor Reliability*

Four aspects of pressure sensor reliability were explored.

1) *Static Load*

The first experiment was to measure the sum of the pressure readings while the subject stood still, i.e., a static load test. The subject started by standing on both feet, then shifted onto their left foot only, then back to both feet, then right foot only, before returning to both feet. Each position was held for approximately 7 seconds, giving a total time of approximately 36 seconds (900 samples). The results are shown in Fig. 3.

The following observations can be made -

1. The total pressure is fairly constant when standing on both feet or just one foot. The total pressure on a foot when moving from both feet to just that one foot roughly doubles, suggesting a linear relationship between total pressure and subject weight.
2. There is +/- 10% variation between samples even when "standing still", due to the ongoing small adjustments the subject makes to maintain their balance, even during a "static" test.
3. When standing on the right foot only, the left foot readings reduce to zero (orange line), but when standing on the left foot only the right foot readings have a low level of noise. If the ECUs were swapped round, then this noise transferred to the other insole, suggesting that it is one of the ECUs that is introducing the noise. The noise was small compared to the overall readings during normal activity (approximately 1-3% relative to the pressure recorded when standing).
4. The scale on the y axis is the sum of the pressure sensor readings. It is not a force because the area over which each sensor measures pressure is not known, although it could possibly be used as a proxy for force (in this case the subject's weight)

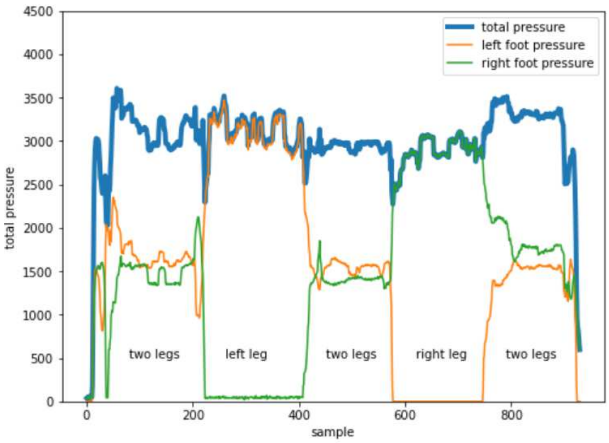

Fig. 3. Static load while standing on one or both feet.

2) *The Influence of Footwear*

The second experiments looked at the influence of footwear on pressure readings. In particular –

- The effect of any layers between the subject's feet and the smart insoles (e.g. socks);
- The effect of the layer between the surface being walked on and the smart insoles (e.g. shoe).

a) *Influence of the Layer between Foot and Smart Insole*

This experiment measured the effect of different types of layer between the subject's feet and the smart insoles. The results are shown in Fig. 4 for the subject standing in three scenarios - bare feet (i.e. no layer between smart insole and feet), feet with socks on and using socks plus the shoe insoles.

It can be seen that the choice of layer(s) has a significant effect. With bare feet total pressure is just under 3000, whereas adding socks causes it to jump to over 4000, and adding the shoe's insoles brings it down to around 3500. It is not clear which of these readings is most accurate because there is no way to convert the total pressure into a force.

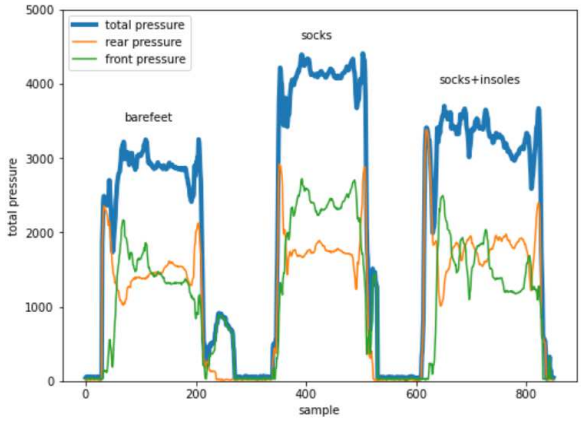

Fig. 4. Total pressure for different layers between foot and smart insole.

b) *Influence of Shoe type*

This experiment measured the effect of different types of shoes (i.e. the layer between the walking surface and the smart insoles). In this case a dynamic test was used, with the subject walking for 100 strides in three different types of shoes, and on three different surfaces. The average total pressure across all 16 pressure sensors was calculated for each scenario. The results are shown in Fig. 5. It can be seen that while the total pressure is fairly consistent across different surfaces, it does vary by up to a third between different types of shoe.

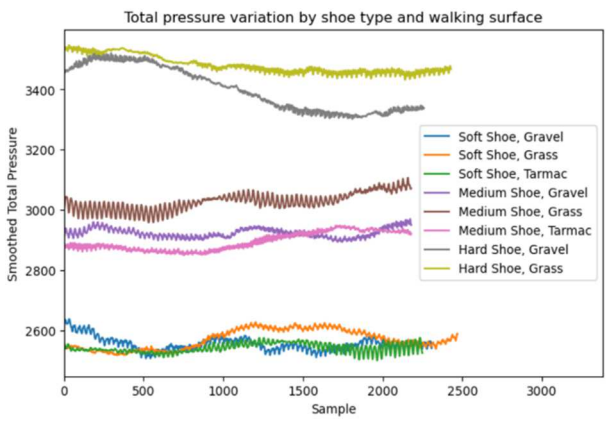

Fig. 5. Average total pressure for different shoes and walking surfaces measured over 100 strides (approximately 2500 samples) and smoothed.

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3) *Smart Insole Position within Shoe*

The third experiment assessed how the position of the insole within the shoe affected the measurements obtained. Two sets of measurements were taken – one with the insoles pushed fully forward in the shoes and one with the insoles positioned at the back of the shoes, a difference of about 3 centimetres (Fig. 6). Each plot shows 5 strides and only the left foot is shown for clarity. The blue line is the total pressure from the left foot, the green line is the sum of the 3 rear sensors and the orange line is the sum of the 5 forward sensors. The figures at the top of each plot show the average pressure for each of the three lines over the 5 strides. It can be seen that -

1. The average total pressure only differs by about 2% between the two plots, but when the insole is pushed forward the average front pressure is approximately 6% higher than when the insole is at the back, and the average rear pressure is 15% lower.
2. The distribution of the pressure within the strides looks visibly different.

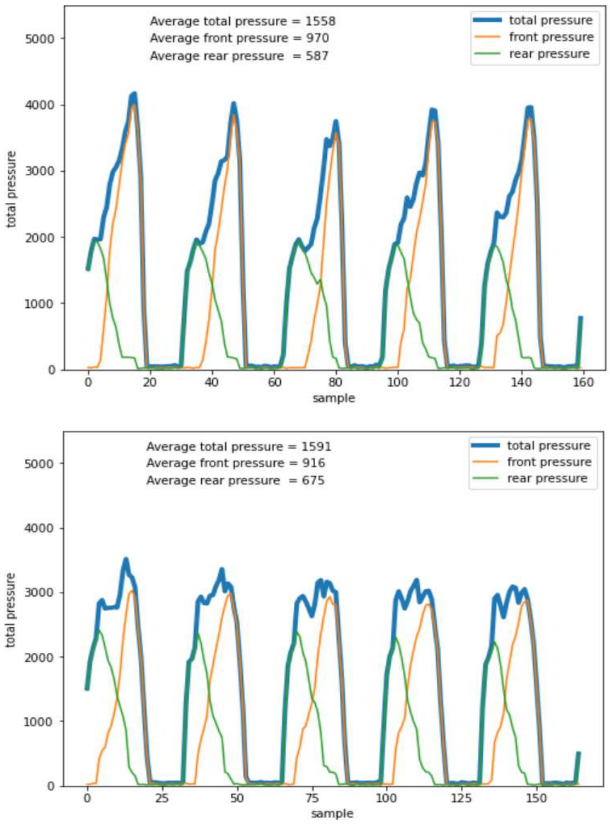

Fig. 6. Variation in total pressure for left foot, with the smart insoles pushed fully forward in the shoes (top) and fully back in the shoes (bottom)

4) *Preload*

The fourth experiment investigated whether tightly tying the shoes introduced any "pre-load" on the pressure sensors. A pair of cycling shoes was used with "BOA lacing" (i.e. lacing where the tightness can be quickly and easily adjusted using a dial). The subject sat with their feet hanging loosely, and pressure was measured (Fig. 7) when the shoes were loose (shoe open), normally fastened, tightly fastened and finally the subject then stood up and sat down again. It can be seen that the tightness of the shoe fasteners had no measurable effect on the pressure measurements. (The low level reading on the left insole is due to the noise introduced by the ECU, as noted earlier – it did not change as the shoes were tightened.)

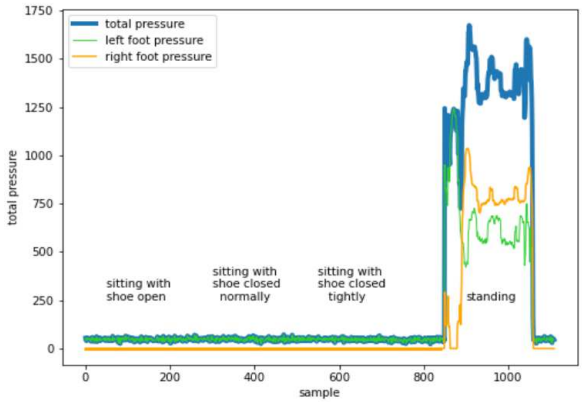

Fig. 7. Total static pressure for different shoe tightness.

B. *Inertial sensor reliability*

One aspect of inertial sensor reliability was explored.

1) *Inertial Sensor Positioning*

This experiment investigated whether the positioning of the ECU on the side of the shoe affected the measurements. Samples were recorded with the ECU attached vertically (as intended), tilted forward by approximately 20 degrees and tilted back by approximately 20 degrees. Note - the ECU is not intended to be used at these angles, but it was found in practice that the clip attaching the ECU to the shoe was quite loose and it was easy to accidentally knock the ECU out of position, or for the ECU to slip out of position while walking. The results for the Magnetometer Y and Z readings (on the right foot, normalised), recorded over 3 strides (approximately 90 samples), are shown in Fig. 8.

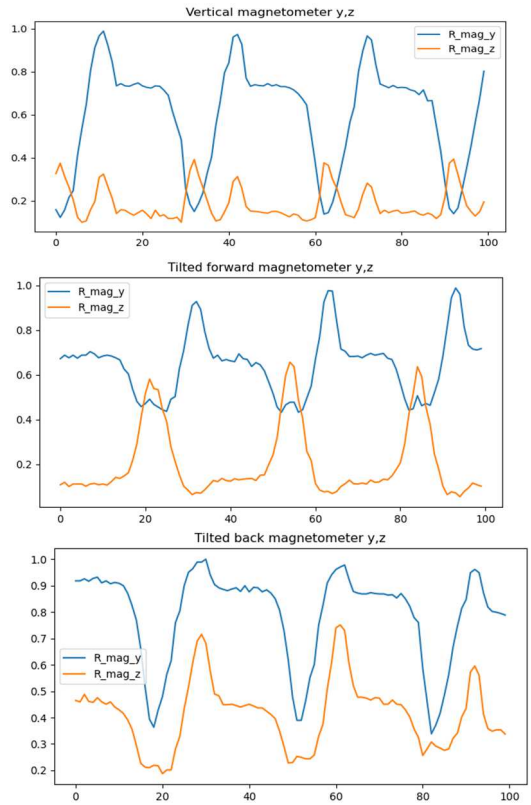

Fig. 8. Magnetometer readings (y and z on the right foot), recorded over three strides showing the impact of ECU positioning (top with ECU correctly positioned vertically, middle with ECU tilted forward and bottom with ECU tilted back. Note the ECU is not intended to be used tilted but can get accidentally knocked or slip during use.)

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It can be seen that when the ECU was vertical the Y reading had a very clear pattern with a strong rising section at the start that could be used for stride detection. But when the ECU was tilted forward this pattern became significantly less prominent, and when the ECU was tilted back, the Z reading started to take on the stronger pattern. This suggests that both the size and pattern of the inertial sensor readings are sensitive to accurate, consistent positioning of the ECUs.

A similar pattern was seen on the Accelerometer readings, but was less discernible on the Gyroscope readings due noise.

C. *Overall system reliability*

Two aspects of overall system reliability were explored.

1) *Drift*

It was noticed that the readings from the left and right feet when the subject was walking normally were not always "half a stride out of sync" as would be expected, see Fig. 9.

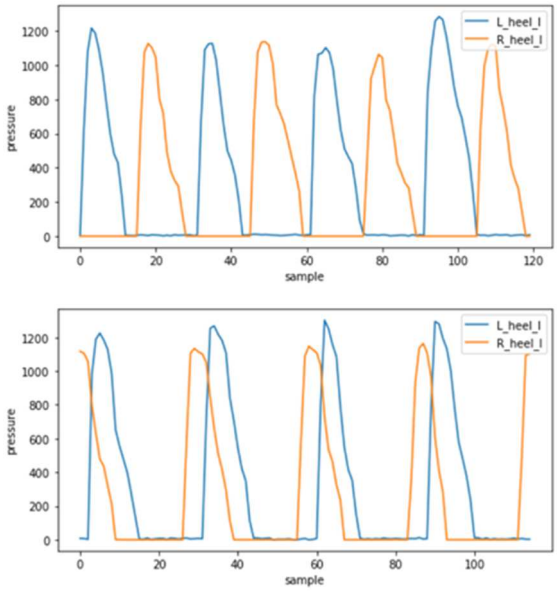

Fig. 9. Plots of four strides, showing drift. In the top plot the left and right heels are half a stride out of sync as expected. In the bottom plot the right heel has drifted out and is about 10 samples later than might be expected.

The measurements from the left and right feet are recorded separately (by two different ECUs). A signal is sent from a controlling device (in this case an app running on an android phone) to each ECU to tell it to start recording, and subsequently stop recording. If multiple readings were taken during a session (i.e. multiple sets of "start and stop" requests were issued), then the "frame counter" gap between the two sets of readings grew larger with each "start and stop". This drift is illustrated in Fig. 10, where each line represents a session (on a particular date), and each dot represents the start or end of a set of readings.

It is possible to try and correct this drift by shifting the readings for one foot relative to the other, but it is difficult to do this with certainty and it could obscure or remove some genuine aspect of the walking, e.g. limping.

2) *Sensor Stickiness*

The second experiment looked at an intermittent problem where some of the pressure sensors seemed to "stick", Fig. 11. In the top plot, the left foot and right foot heel sensors are similar, as would be expected. In the bottom plot, the left foot readings seem to "get stuck" at around 600 and do not return

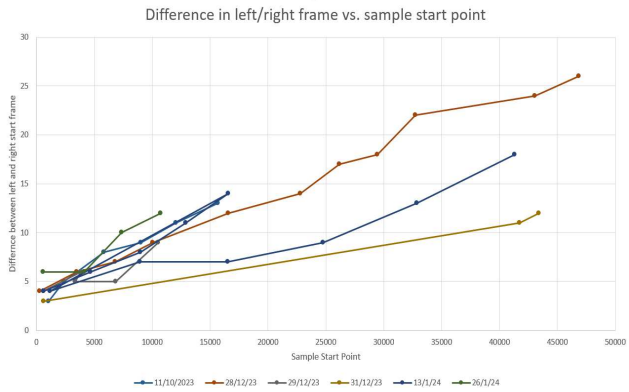

Fig. 10. Plot showing the size of drift between left and right insoles, versus the sample start point within a session.

to zero when the foot leaves the surface. This problem is easily detected visually in the raw data collected from the sensors, but is much harder to detect if the data is normalised (as often happens if the data is used in Machine Learning applications) because the range of the raw data (600 to 1400 in this case) will be mapped to the range 0 to 1, just like the other sensors. It was also intermittent.

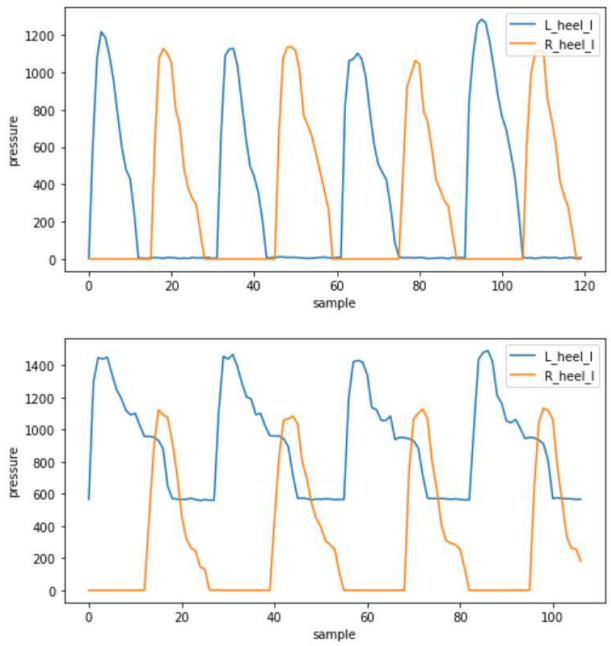

Fig. 11. Plots of four strides. In the top plot the left and right heel_l sensor recordings are as expected. In the lower plot, the left heel_l sensor seems to get "stuck" at 600 rather than returning to zero.

V. Implications

The experiments and results described in the previous section have a number of implications for the use of smart insoles in reliable HAR applications –

1. The pressure sensor measurements cannot be directly converted into force values. If force values are needed, then some sort of calibration will be needed, e.g., using the subject's weight. Also, static pressure/ force measurements are difficult to obtain due to the dynamic way in which human subjects make continual small adjustments to maintain balance.
2. The selection of shoe, sock and insoles can have a big influence on the pressure measurements. If results are to be compared between sessions and/or between

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different subjects, then identical combinations of shoe, sock and insole should be used.

3. The position of the smart insole within the shoe can have an impact on pressure measurements. If an application depends on the size or pattern of pressure measurements then consistent positioning of the insole within the shoe is critical.
4. Preload (i.e. how tightly the shoe is fastened) was not found to affect pressure measurements.
5. Some inertial sensor measurements were found to be noisy. This can be reduced by pre-processing (e.g., filtering) but this needs to be traded off against other factors (e.g., processing overhead and delay).
6. Consistent positioning of the ECUs (both during a session, and across different sessions) is important for applications that depend on the size and pattern of inertial sensor measurements.
7. Drift was seen between the left and right foot measurements. If the relative timing of the left and right measurements is important for an application, then some steps to detect and remove or minimise this should be considered. (Note IEE [7] state that they are looking at taking steps to improve ECU synchronisation in a future release).
8. Some intermittent sensor issues were seen (e.g., pressure sensor "stickiness"). It would be worth using pre-processing to try and detect and flag such issues.

In summary smart insoles have strengths and weaknesses. They are well suited to some HAR applications, whereas care needs to be taken when using them in other scenarios.

Examples of scenarios to which smart insoles are well suited include -

1. Scenarios where a change in force needs to be measured, but the absolute size of the force is less important. For example, if a subject was required to complete an exercise where they stood on their heels, then rolled forward onto their toes, and then back again, an app could illustrate the target pattern of pressure distribution to aim for, and the actual pressure distribution being achieved by the subject. This would not need accurate or consistent pressure measurements, but only the overall pattern of change.
2. Stride times can be measured by looking for a sensor with a clear, repeated pattern, and picking a point where the sensor reading is changing rapidly at the same point in each stride. Stride length is useful in many HAR applications.

Examples of scenarios where use of smart insoles requires care include -

1. Any application that requires an absolute measure of pressure or force, or which is looking for small changes in the measurements over time. For example, trying to measure improvements to right/left balance over time would require careful attention to aspects such as the use of the same shoes, insoles and socks, and to the position of the insoles. It might be worth doing some sort of "calibration" prior to each session, possibly using the subject's weight as a known measurement.
2. Any application that requires accurate inertial measurements. These require accurate mounting of the inertial sensors, and care that the mounting is not disturbed during the session.

VI. Conclusions and Future Work

This paper has identified several challenges with the reliability of data collected using smart insoles. But with careful design and careful use of the smart insoles in execution, useful results can be obtained.

Further work is needed in this area, for example -

1. This study only looked at one type of smart insoles. The reliability of smart insoles from a wider variety of suppliers should be assessed.
2. This study only looked at the measurements from a single subject. The results obtained across multiple subjects should be assessed.
3. This study only looked at reliability at a single point in time. Variations in reliability over time (e.g., degradation or drift) should be assessed.
4. All experiments in this study were carried out with a sampling frequency of 25Hz. Faster sampling speeds are needed in many applications, and the impact of this on reliability should be assessed.
5. This study only looked at measurements taken outdoors. The impact of location (e.g. magnetometer readings inside a steel framed building) should be assessed.
6. The effect on smart insole measurements of walking or running on different surfaces should be investigated.

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